



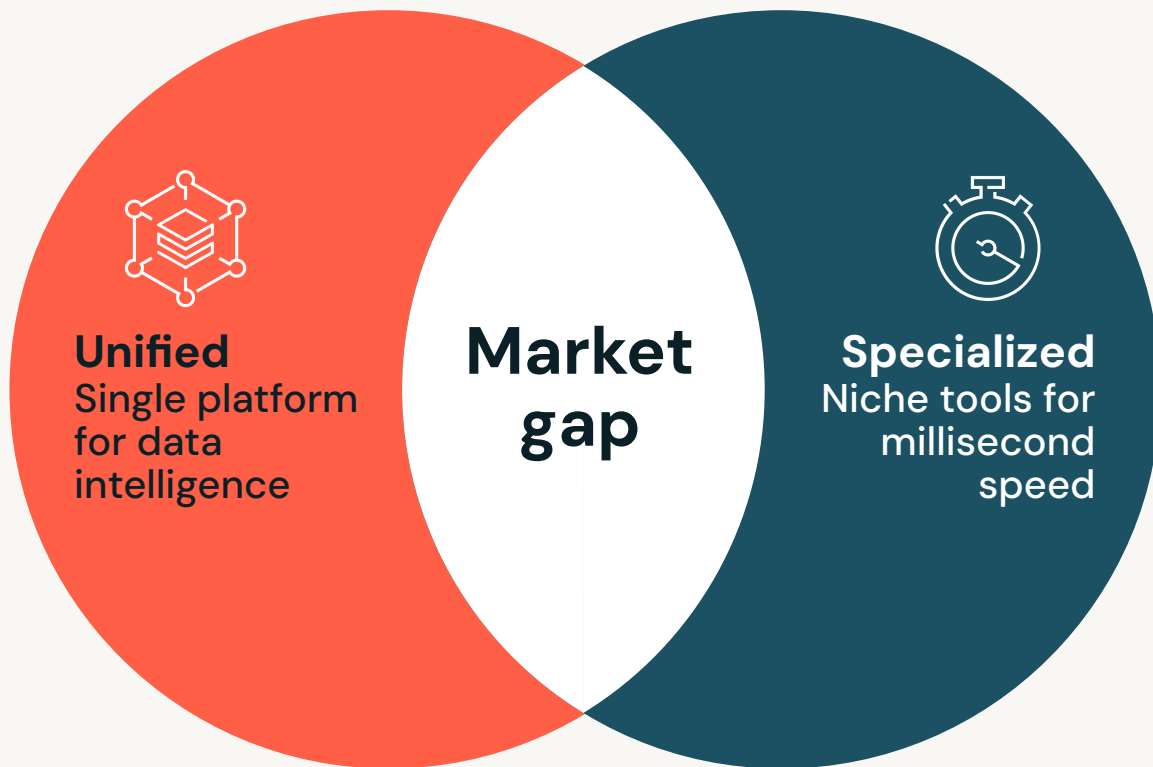
Lakehouse//RT

Millisecond performance at
massive scale — without
leaving your lakehouse

July 2026



But no one has solved both speed and unification



Today, companies rely on **side stacks** for real-time needs.

- ✗ Extra costs & overhead
- ✗ Data duplication
- ✗ System sprawl
- ✗ Proprietary formats
- ✗ Pipeline maintenance
- ✗ Broken governance

 ClickHouse

 pinot

 + a b l e a u

 Power BI

 druid

star⁺tree

 DuckDB

 imply

 BigQuery

 snowflake



But Databricks delivers a unified data platform



Agent Bricks
Production AI agents



AI/BI
Agentic business intelligence



Custom Apps
Secure data and AI apps

And more...



AI with Enterprise Context



Genie



Lakeflow
Ingest, ETL, streaming



Lakehouse
Data warehousing



Lakebase
Serverless Postgres



Unified Governance



Unity Catalog



Open Formats



Postgres



DELTA LAKE

ICEBERG



Lakehouse//RT, a real-time lakehouse

Millisecond performance at massive scale — directly on your lake



Speed at scale

Get millisecond performance for real-time workloads where high concurrency is needed



Unified platform

Rely on a single, agile platform — no duplication, no extra pipelines, no proprietary formats



Fully governed

Protect your data and business logic with consistent governance and a single source of truth



Lakehouse//RT is a new warehouse type



Agent Bricks
Production AI agents



AI/BI
Agentic business
intelligence



Custom Apps
Secure data and AI apps

And more...

Select Lakehouse//RT from the Warehouse Picker. That's it.

Last edit was now



▶ Run all



⚡ Lakehouse//RT Warehouse ▼

Schedule

Share



⚡ Lakehouse//RT

Recent



Serverless



Serverless GPU **Beta**

Recent resources



0 - Shared SQL Warehouse **SQL**

X-Large

Active resources



[DCS vNext] OAI Demo Cluster

17.3 LTS • 1-2 workers



Rebuilt to shatter the scale-latency tradeoff

A rebuilt, native engine...

Built for modern hardware.
No JVM overhead. Caching everything

...that speaks Photon...

Passes data and plans directly to Photon for a seamless, end-to-end vectorized pipeline

...to power real-time workloads.

Drives agentic use cases, BI serving, operational analytics, and workloads where speed is critical



...powered by



Reyden

Separate serving stack

- ✗ Can't run queries >5s
- ✗ Copy to proprietary formats
- ✗ Can't run complex joins
- ✗ Costs spike with scale
- ✗ Fragmented governance

Lakehouse//RT powered by Reyden

- ✓ Run queries small and large
- ✓ Reads open table formats
- ✓ Handles complex joins
- ✓ Autoscales for lower TCO
- ✓ Governed by Unity Catalog

When to use Lakebase, DBSQL, or Lakehouse//RT?

One platform, one copy of data, one governance model. Three engines for three jobs.

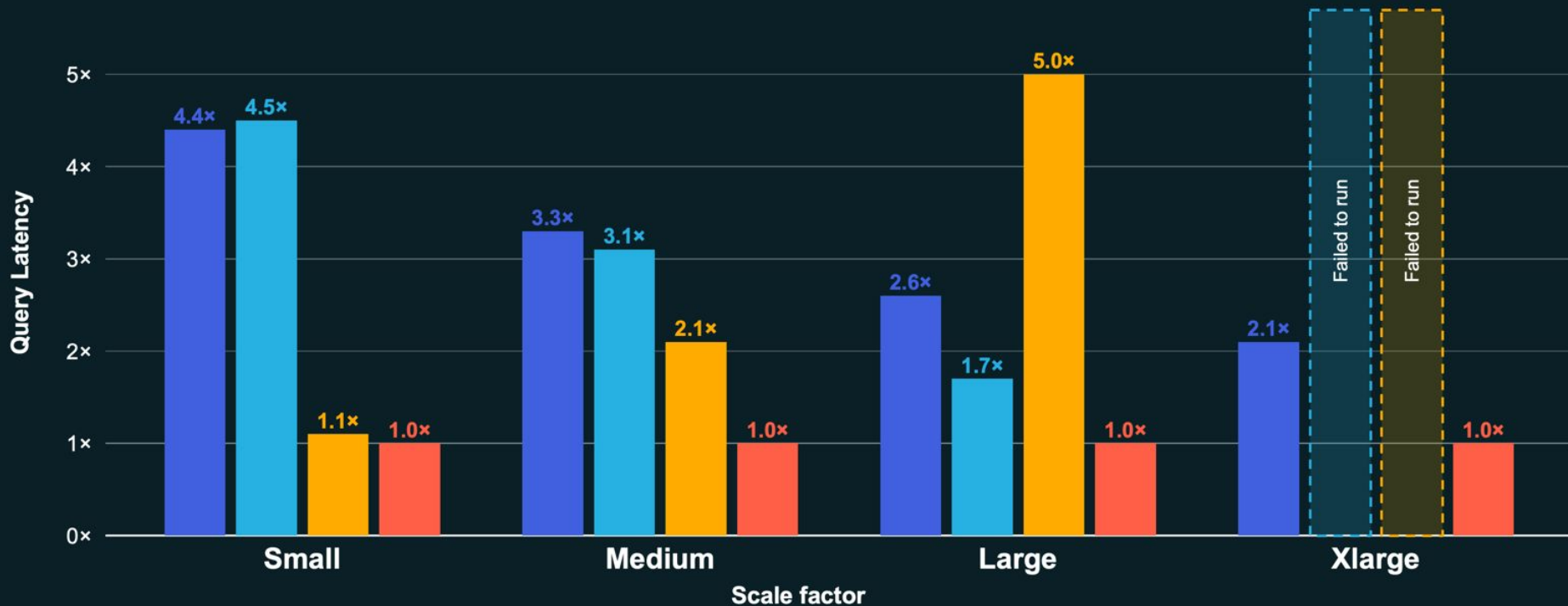
	Lakebase	DBSQL	Lakehouse//RT
What it is	Managed Postgres (OLTP)	Serverless data lakehouse	Real-time lakehouse
Built for	Transactions and app state	Analytics and BI at scale	Real-time analytics at scale
Query shape	Point reads / writes, single-row	Complex scans, joins, aggregations, DML	Read only analytics, no large scans
Data scale	2-16TB	Unbounded	Unbounded dataset, bounded per-query (runtime < 2mins)
Latency	Milliseconds	Seconds	Milliseconds
Concurrency	10K+ QPS, up to 1,000 connections	~10 queries/cluster, autoscales to 100s	2K-4K QPS
Data	PG format, synced to lake	Open table formats (Delta/Iceberg)	Open table format (Delta/Iceberg)

Use when	App backends, agent state, system of record	Dashboards, ETL, exploration, heavy analytical queries	Real-time analytics, BI serving, ops analytics
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Fast on the largest data sets

Benchmark based on TPC-H. Lower is better

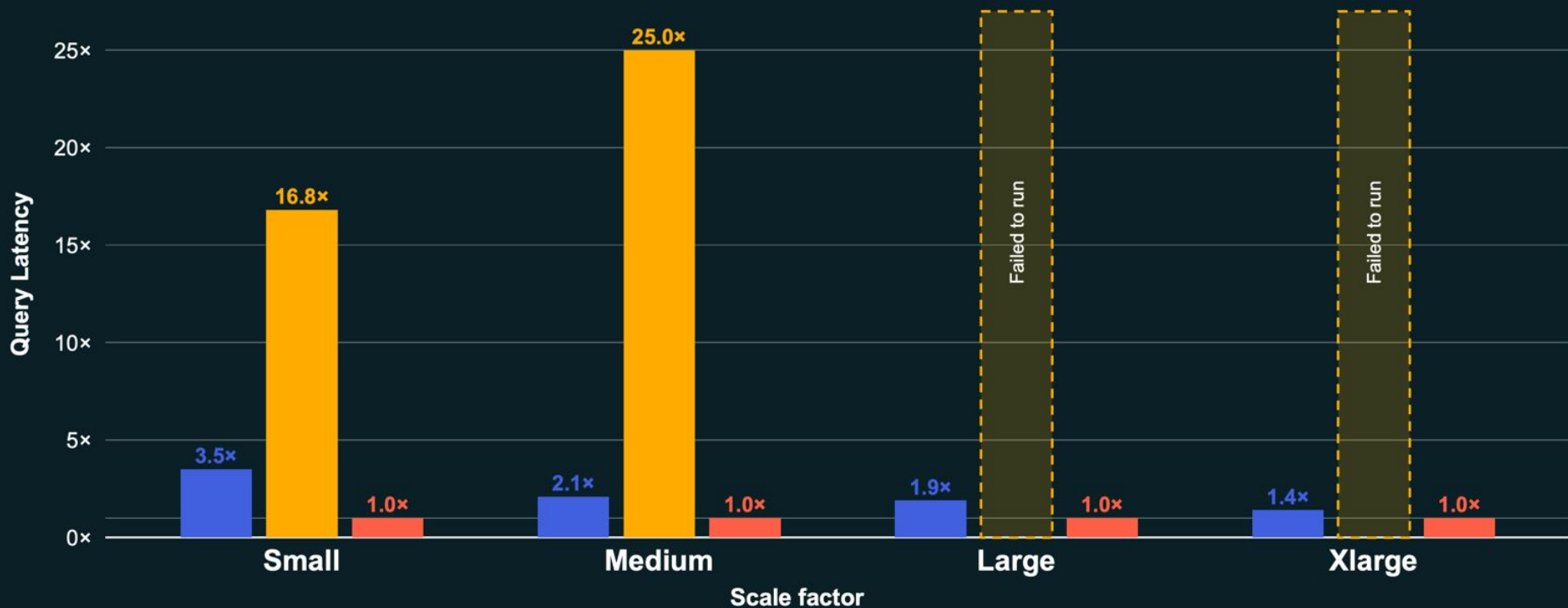
Vendor 1 (Gen 2) Vendor 1 (Interactive) Vendor 2 Reyden



Fast on the hardest analytical queries

Benchmark based on TPCDS. Lower is better

Vendor 1 (Gen 2) Vendor 2 Reyden



Lakehouse//RT is a new warehouse option

Expanding compute options for diverse workloads



Lakehouse

SQL Classic

Traditional compute
for SQL

SQL Pro

Faster, upgraded with
Predictive IO

SQL Serverless

Fully managed, elastic,
on-demand

Lakehouse RT

High concurrency, low
latency, for millisecond
speed at scale



Vectorized query
engines

Query
acceleration
services

RDBMS

MPP databases

What makes our **real-time lakehouse**
better than other specialized engines?

Distributed OLAP
datastores

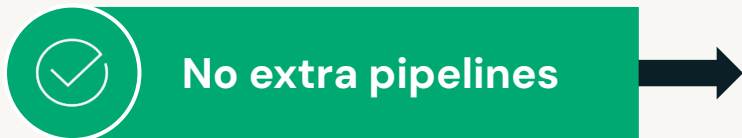
Specialized
databases

In-memory
engines

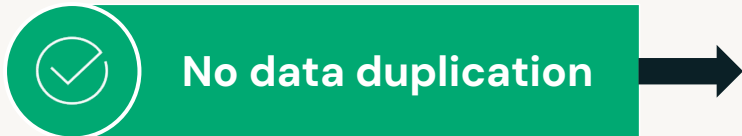


Simple, sustainable speed at scale

A streamlined architecture lets agents work at top-speed with the best data



Eliminate the risk and cost of maintenance and extra movement



Query directly at the source – not subsets – for maximum freshness



Standardize on data you own, in open formats, without lock-in



Let agents move at top speed with complete, full-context data



Running on open data & unified governance

One single source of truth and control across all your data with Unity Catalog



Single source

Zero copies, zero drift



Unified policy

One rule, everywhere



Complete lineage

Auditability for agents

Unleash your most ambitious AI use cases with the confidence of a single, unified control panel and data you own on open formats



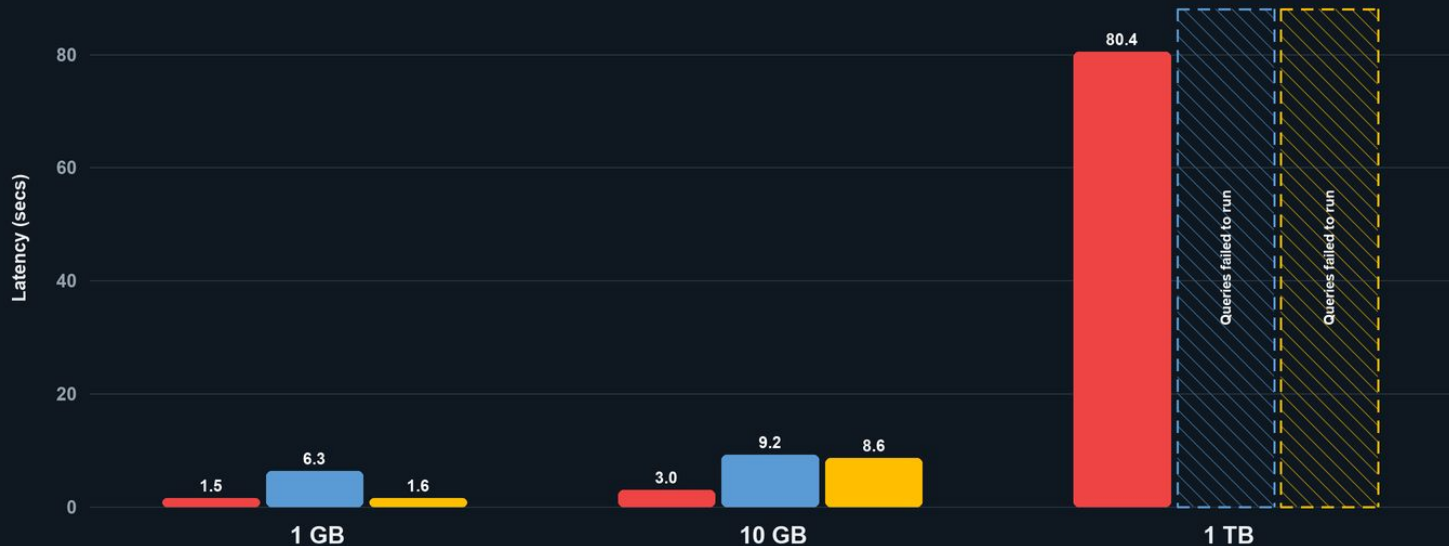
What if you could get millisecond
performance at massive scale –
directly on your lakehouse?



Scaling from small to big data on a single copy of your data

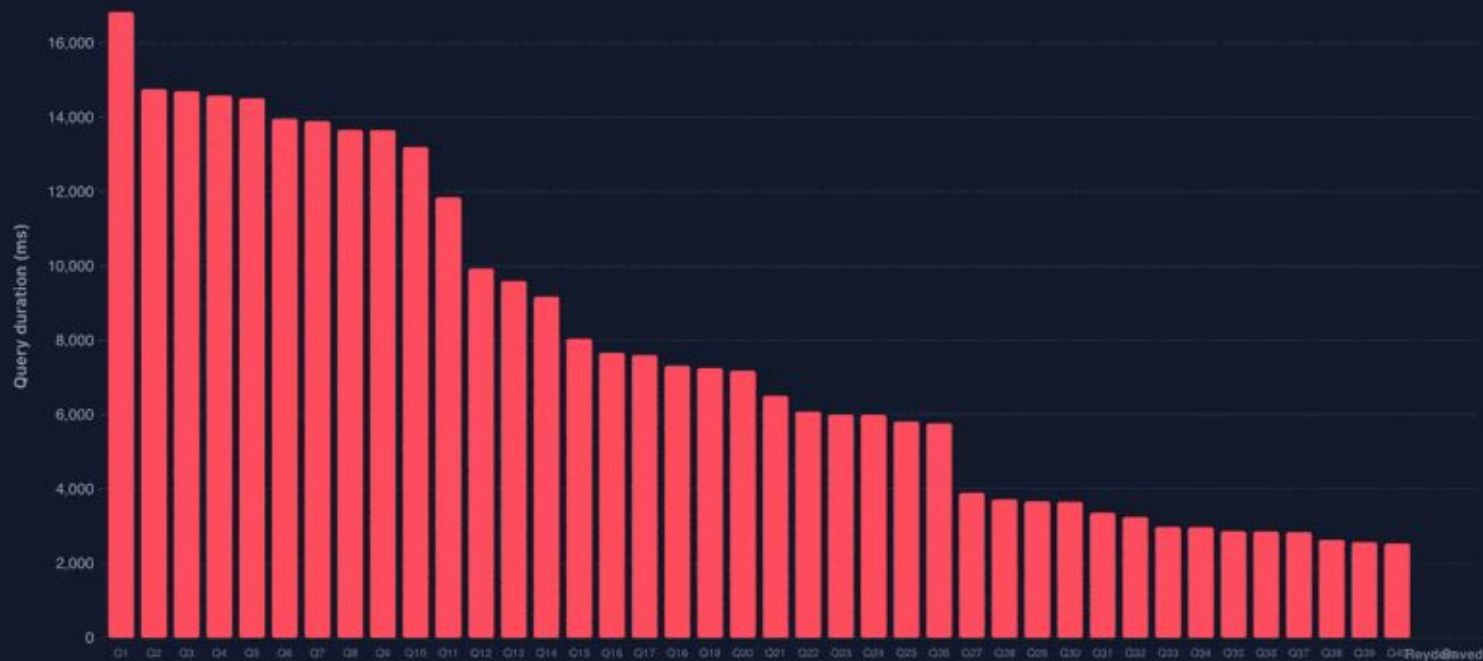
Lower is better

● Lakehouse//RT ● Vendor 1 ● Vendor 2 – Interactive



A real customer workload

These are Zillow's **40 slowest queries**.



So... what's
the **catch?**

No catch! Just:



One copy of your data



No migration



Same price as DBSQL



With Lakehouse//RT

we can now go for ALL
performance sensitive
usecases



Operational monitoring

Reporting, dashboards, BI, & exploration



Customer facing & agent applications

Agentic applications, agents, & APIs



Real-time analytics

User-facing, event-driven decisions



Product analytics

User behavior & clickstream



Threat detection & security

SIEM & security webhooks



Observability & monitoring

Operational logs, metrics, & traces



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5 key takeaways

1

Start with Managed Tables

Predictive optimization, auto clustering, metadata caching

2

Optimize data scans

Use the right liquid clustering keys and enroll in the **Gated Preview (Aug 2026) and Secondary Indexing PrPr (H1, 2027)** so queries scan less data.

3

Right size the cluster

Match compute to the workload instead of over- or under-provisioning

4

Use OPTIMIZE NOW PrPr for POCs

Immediately kick off PO and Auto clustering

Frequently asked questions



What workloads are best suited for Lakehouse//RT?

Lakehouse//RT is designed for analytical workloads where millisecond responses at high concurrency are needed, such as BI serving or operational analytics.



What BI tools and connectors work with Lakehouse//RT?

Lakehouse//RT supports Databricks AI/BI as well as the SQL Exec API (SEA), PowerBI, and QuickSight – with more planned.



Do I need to change my underlying data formats or rebuild my existing pipelines?

Lakehouse//RT works with Iceberg, Delta, and Parquet formats. Data modeling best practices still need to be followed to get the best performance.



What's the difference between Lakehouse and Lakehouse//RT?

Lakehouse (formerly Databricks SQL) is a cloud data warehouse built on lakehouse architecture. Just as SQL Serverless is a deployment option you might use today, Lakehouse//RT is a new warehouse option built for real-time needs.



What's the difference between Lakehouse//RT and Lakebase?

Lakehouse//RT is a high-speed engine built for low latency and high concurrency — running massive OLAP queries straight out of your data lake — while Lakebase is an OLTP database meant for pinpoint lookups and updating app data.



What's the difference between Lakehouse//RT and Zerobus Ingest?

Zerobus Ingest is used for ingestion; Lakehouse//RT is for real-time queries or 'reads'. In use cases with E2E latency requirements where data must be fresh, such as observability, Zerobus Ingest + Lakehouse//RT is the right combination.



Reyden – “Caveats”

1

It is not "out of the box" yet.

~35% of POCs needed manual tuning pre-Lakehouse//RT. Massive perf gains only show on managed tables, PO, and auto-liquid on by default.

2

Every query runs inside hard guardrails.

2-minute runtime cap, 300GB shuffle and 30GB spill limits. Ideal fit for queries than need to run in Milliseconds to seconds.

3

It only reads. No writes, no ETL.

No write, merge, update, or delete. Writes land through your existing pipeline and Reyden serves the reads. Data engineering and transformations stay on DBSQL and Jobs.

4

The connector ecosystem is still thin.

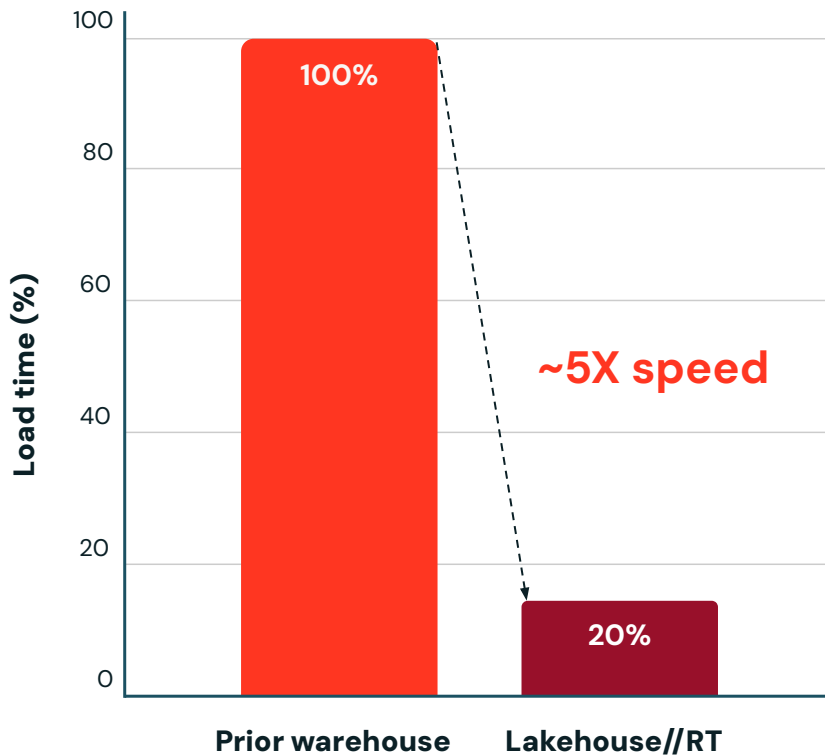
SQL Exec API only. AI/BI and Metric Views work today; PowerBI, JDBC, and the Go connector are still WIP, and QuickSight needs an AWS driver upgrade. If a tool is not on SEA yet, the customer waits.



Threat lookup requires consistently low latency, even as usage scales across users and agents.

What we're seeing with Lakehouse//RT is millisecond performance on live data with **5x improvement** in response time, which creates a path to run those workloads **on our lakehouse** instead of maintaining a separate serving system.

Chris Kopek
Head of Data Platforms



Magnite



With Lakehouse//RT, we're seeing **sub-200 millisecond performance** on our core dashboard queries, consistently.

Being able to achieve that while keeping everything governed inside our own data lake **massively reduces the complexity** of managing our data pipeline and servicing of consumer applications.



Kayvon Raphael

Senior Director, Engineering



Thank you

